4 Vector Spaces

1

4.1 Vectors in R^n

2

Objectives

- Represent a vector as a directed line segment.
- Perform basic vector operations in R^2 and represent them graphically.
- Perform basic vector operations in R^n.
- Write a vector as a linear combination of other vectors.

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Vectors In The Plane

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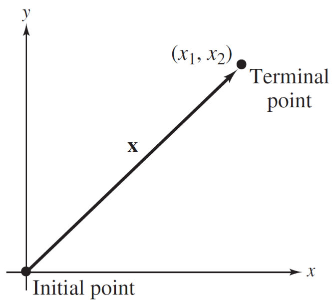
Vectors In The Plane (1 of 2)

Geometrically, a **vector in the plane** is represented by a **directed line segment** with its **initial point** at the origin and its **terminal point** at (x_1, x_2), as shown below.



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Vectors In The Plane (2 of 2)

The same **ordered pair** used to represent its terminal point also represents the vector. That is, $\mathbf{x} = (x_1, x_2)$. The coordinates x_1 and x_2 are the **components** of the vector **x**.

Two vectors in the plane $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ are **equal** if and only if

$$u_1 = v_1 \quad \text{and} \quad u_2 = v_2.$$

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Example 1 – Vectors in the Plane (1 of 2)

a. To represent **u** = (2, 3), draw a directed line segment from the origin to the point (2, 3), as shown in Figure 4.1(a).

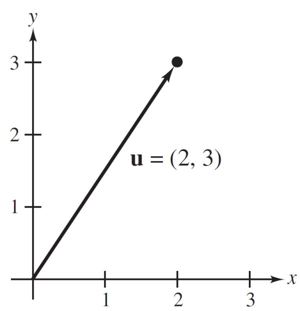

Figure 4.1(a)

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Example 1 – Vectors in the Plane (2 of 2)

b. To represent **v** = (−1, 2), draw a directed line segment from the origin to the point (−1, 2), as shown in Figure 4.1(b).

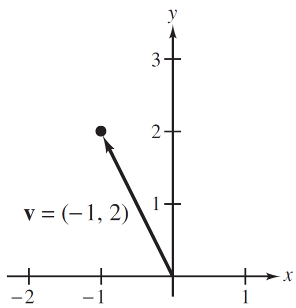

Figure 4.1(b)

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Vector Operations

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Vector Operations (1 of 6)

One basic vector operation is **vector addition.** To add two vectors in the plane, add their corresponding components. That is, the **sum** of **u** and **v** is the vector

$$\mathbf{u} + \mathbf{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2).$$

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Vector Operations (2 of 6)

Geometrically, the sum of two vectors in the plane can be represented by the diagonal of a parallelogram having **u** and **v** as its adjacent sides, as shown in Figure 4.2.

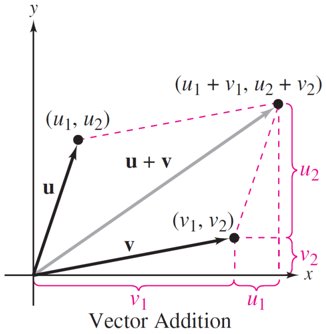

Figure 4.2

Example 2 – Adding Two Vectors in the Plane

Find each vector sum **u** + **v**.

a. **u** = (1, 4), **v** = (2, −2)

b. **u** = (3, −2), **v** = (−3, 2)

c. **u** = (2, 1), **v** = (0, 0)

Solution:

a. **u** + **v** = (1, 4) + (2, −2) = (3, 2)

b. **u** + **v** = (3, −2) + (−3, 2) = (0, 0) = **0**

c. **u** + **v** = (2, 1) + (0, 0) = (2, 1)

Example 2 – Solution

Figure 4.3 shows a graphical representation of each sum.

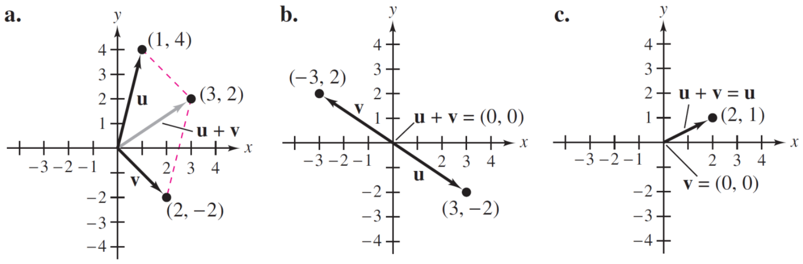

Figure 4.3

Vector Operations (3 of 6)

Another basic vector operation is **scalar multiplication.** To multiply a vector **v** by a scalar c, multiply each of the components of **v** by c. That is,

$$c\mathbf{v} = c(v_1, v_2) = (cv_1, cv_2).$$

Example 3 – Operations with Vectors in the Plane

Let **v** = (−2, 5) and **u** = (3, 4). Perform each vector operation.

a. $\frac{1}{2}\mathbf{v}$ **b.** $\mathbf{u} - \mathbf{v}$ **c.** $\frac{1}{2}\mathbf{v} + \mathbf{u}$

Solution:

a. **v** = (−2, 5), so $\frac{1}{2}\mathbf{v} = \left(\frac{1}{2}(-2), \frac{1}{2}(5)\right) = \left(-1, \frac{5}{2}\right)$.

b. By the definition of vector subtraction,

u − **v** = (3 − (−2), 4 − 5) = (5, −1).

Example 3 – Solution

c. Using the result of part (a),

$$\frac{1}{2}\mathbf{v} + \mathbf{u} = \left(-1, \frac{5}{2}\right) + (3, 4) = \left(2, \frac{13}{2}\right).$$

Figure 4.5 shows a graphical representation of these vector operations.

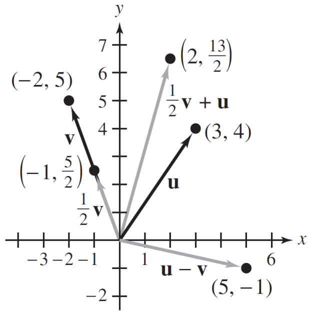

Figure 4.5

Vector Operations (4 of 6)

Theorem 4.1 Properties of Vector Addition and Scalar Multiplication in the Plane

Let **u**, **v**, and **w** be vectors in the plane, and let *c* and *d* be scalars.

1. **u** + **v** is a vector in the plane.	Closure under addition
2. **u** + **v** = **v** + **u**	Commutative property of addition
3. (**u** + **v**) + **w** = **u** + (**v** + **w**)	Associative property of addition

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Vector Operations (5 of 6)

Theorem 4.1 Properties of Vector Addition and Scalar Multiplication in the Plane

4. **u** + **0** = **u**	Additive identity property
5. **u** + (−**u**) = **0**	Additive inverse property
6. *c***u** is a vector in the plane.	Closure under scalar multiplication
7. *c*(**u** + **v**) = *c***u** + *c***v**	Distributive property
8. (*c* + *d*)**u** = *c***u** + *d***u**	Distributive property

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Vector Operations (6 of 6)

Theorem 4.1 Properties of Vector Addition and Scalar Multiplication in the Plane

9. $c(d\mathbf{u}) = (cd)\mathbf{u}$	Associative property of multiplication
10. $1(\mathbf{u}) = \mathbf{u}$	Multiplicative identity property

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Vectors in R^n

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Vectors in R^n (1 of 8)

An **ordered *n*-tuple** represents a vector in *n*-space. For instance, an ordered triple has the form (x_1, x_2, x_3), an ordered quadruple has the form (x_1, x_2, x_3, x_4), and a general ordered *n*-tuple has the form $(x_1, x_2, x_3, \ldots, x_n)$. The set of all *n*-tuples is ***n*-space** and is denoted by R^n.

Vectors in R^n (2 of 8)

R^1 = 1-space = set of all real numbers

R^2 = 2-space = set of all ordered pairs of real numbers

R^3 = 3-space = set of all ordered triples of real numbers

$\vdots$

R^n = *n*-space = set of all ordered *n*-tuples of real numbers

An *n*-tuple $(x_1, x_2, x_3, \ldots, x_n)$ can be viewed as a **point** in R^n with the x_i's as its coordinates, or as a **vector** $x = (x_1, x_2, x_3, \ldots, x_n)$ with the x_i's as its components.

Vectors in R^n (3 of 8)

Definitions of Vector addition and scalar Multiplication in R^n

Let **u** = $(u_1, u_2, u_3, \ldots, u_n)$ and **v** = $(v_1, v_2, v_3, \ldots, v_n)$ be vectors in R^n and let c be a real number. The sum of **u** and **v** is the vector

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3, \ldots, u_n + v_n)$$

and the **scalar multiple** of **u** by c is the vector

$$c\mathbf{u} = (cu_1, cu_2, cu_3, \ldots, cu_n).$$

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Example 4 – Vector Operations in R^3

Let **u** = (−1, 0, 1) and **v** = (2, −1, 5) in R^3. Perform each vector operation.

a. **u** + **v** **b.** 2**u** **c.** **v** − 2**u**

Solution:

a. To add two vectors, add their corresponding components.

$$\mathbf{u} + \mathbf{v} = (-1, 0, 1) + (2, -1, 5) = (1, -1, 6)$$

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Example 4 – Solution (1 of 2)

b. To multiply a vector by a scalar, multiply each component by the scalar.

$2\mathbf{u} = 2(-1, 0, 1) = (-2, 0, 2)$

c. Using the result of part (b),

$$\mathbf{v} - 2\mathbf{u} = (2, -1, 5) - (-2, 0, 2)$$
$$= (4, -1, 3).$$

Example 4 – Solution (2 of 2)

Figure 4.6 shows a graphical representation of vector operations in R^3.

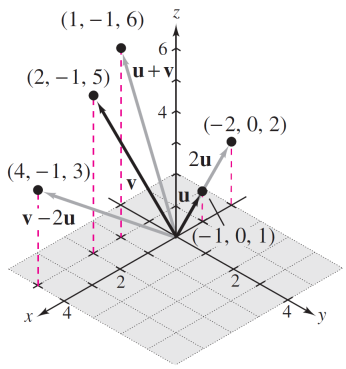

Figure 4.6

Vectors in R^n (4 of 8)

Theorem 4.2 Properties of Vector Addition and Scalar Multiplication in R^n

Let **u**, **v**, and **w** be vectors in R^n, and let *c* and *d* be scalars.

1. **u** + **v** is a vector in R^n. — Closure under addition

2. **u** + **v** = **v** + **u** — Commutative property of addition

3. (**u** + **v**) + **w** = **u** + (**v** + **w**) — Associative property of addition

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Vectors in R^n (5 of 8)

Theorem 4.2 Properties of Vector Addition and Scalar Multiplication in R^n

4. **u** + **0** = **u** — Additive identity property

5. **u** + (−**u**) = **0** — Additive inverse property

6. *c***u** is a vector in R^n. — Closure under scalar multiplication

7. *c*(**u** + **v**) = *c***u** + *c***v** — Distributive property

8. (*c* + *d*)**u** = *c***u** + *d***u** — Distributive property

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Vectors in R^n (6 of 8)

Theorem 4.2 Properties of Vector Addition and Scalar Multiplication in R^n

9. $c(d\mathbf{u}) = (cd)\mathbf{u}$	Associative property of multiplication
10. $1(\mathbf{u}) = \mathbf{u}$	Multiplicative identity property

Example 5 – Vector operations in R^4

Let $\mathbf{u} = (2, -1, 5, 0)$, $\mathbf{v} = (4, 3, 1, -1)$, and $\mathbf{w} = (-6, 2, 0, 3)$ be vectors in R^4. Find $\mathbf{x}$ using each equation.

a. $\mathbf{x} = 2\mathbf{u} - (\mathbf{v} + 3\mathbf{w})$

b. $3(\mathbf{x} + \mathbf{w}) = 2\mathbf{u} - \mathbf{v} + \mathbf{x}$

Solution:

a. Using the properties listed in Theorem 4.2, you have

$$\begin{aligned}\mathbf{x} &= 2\mathbf{u} - (\mathbf{v} + 3\mathbf{w}) \\ &= 2\mathbf{u} - \mathbf{v} - 3\mathbf{w} \\ &= 2(2, -1, 5, 0) - (4, 3, 1, -1) - 3(-6, 2, 0, 3)\end{aligned}$$

Example 5 – Solution (1 of 2)

$= (4, -2, 10, 0) - (4, 3, 1, -1) - (-18, 6, 0, 9)$

$= (4 - 4 + 18, -2 - 3 - 6, 10 - 1 - 0, 0 + 1 - 9)$

$= (18, -11, 9, -8).$

b. Begin by solving for **x**.

$$3(\mathbf{x} + \mathbf{w}) = 2\mathbf{u} - \mathbf{v} + \mathbf{x}$$

$$3\mathbf{x} + 3\mathbf{w} = 2\mathbf{u} - \mathbf{v} + \mathbf{x}$$

$$3\mathbf{x} - \mathbf{x} = 2\mathbf{u} - \mathbf{v} - 3\mathbf{w}$$

$$2\mathbf{x} = 2\mathbf{u} - \mathbf{v} - 3\mathbf{w}$$

$$\mathbf{x} = \tfrac{1}{2}(2\mathbf{u} - \mathbf{v} - 3\mathbf{w})$$

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Example 5 – Solution (2 of 2)

Using the result of part (a),

$$\mathbf{x} = \tfrac{1}{2}(18, -11, 9, -8)$$
$$= \left(9, -\tfrac{11}{2}, \tfrac{9}{2}, -4\right).$$

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Vectors in R^n (7 of 8)

Theorem 4.3 Properties of additive identity and additive inverse

Let **v** be a vector in R^n, and let c be a scalar. Then the properties below are true.

1. The additive identity is unique. That is, if **v** + **u** = **v**, then **u** = **0**.

2. The additive inverse of **v** is unique. That is, if **v** + **u** = **0**, then **u** = −**v**.

3. 0**v** = **0**

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Vectors in R^n (8 of 8)

Theorem 4.3 Properties of additive identity and additive inverse

4. c**0** = **0**

5. If c**v** = **0**, then c = 0 or **v** = **0**.

6. −(−**v**) = **v**

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Linear Combinations of Vectors

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Linear Combinations of Vectors (1 of 1)

An important type of problem in linear algebra involves writing one vector **x** as the sum of scalar multiples of other vectors $\mathbf{v}_1$, $\mathbf{v}_2$, ..., and $\mathbf{v}_n$. That is, for scalars c_1, c_2, ..., c_n,

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n.$$

The vector **x** is called a **linear combination** of the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, ..., and $\mathbf{v}_n$.

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Example 6 – Writing a Vector as a linear combination of other Vectors

Let **x** = (−1, −2, −2), **u** = (0, 1, 4), **v** = (−1, 1, 2), and **w** = (3, 1, 2) in R^3. Find scalars *a*, *b*, and *c* such that

x = *a***u** + *b***v** + *c***w**.

Solution:

Write

$$\overbrace{(-1, -2, -2)}^{\mathbf{x}} = a\overbrace{(0, 1, 4)}^{\mathbf{u}} + b\overbrace{(-1, 1, 2)}^{\mathbf{v}} + c\overbrace{(3, 1, 2)}^{\mathbf{w}}$$
$$= (-b + 3c, a + b + c, 4a + 2b + 2c)$$

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Example 6 – Solution

Equate corresponding components so that they form the system of three linear equations in *a*, *b*, and *c* shown below.

$$-b + 3c = -1 \qquad \text{Equation from first component}$$
$$a + b + c = -2 \qquad \text{Equation from second component}$$
$$4a + 2b + 2c = -2 \qquad \text{Equation from third component}$$

Solve for *a*, *b*, and *c* to get *a* = 1, *b* = −2, and *c* = −1. As a linear combination of **u**, **v**, and **w**,

x = **u** − 2**v** − **w**.

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4 Vector Spaces





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4.2 Vector Spaces

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Objectives

- Define a vector space and recognize some important vector spaces.
- Show that a given set is not a vector space.

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Definition of a Vector Space

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Definition of a Vector Space (1 of 6)

Definition of a Vector Space

Let V be a set on which two operations (**vector addition** and **scalar multiplication**) are defined. If the listed axioms are satisfied for every **u**, **v**, and **w** in V and every scalar (real number) c and d, then V is a **vector space.**

Addition:

1. u + **v** is in V.	Closure under addition
2. u + **v** = **v** + **u**	Commutative property
3. u + (**v** + **w**) = (**u** + **v**) + **w**	Associative property

Definition of a Vector Space (2 of 6)

Definition of a Vector Space

4. V has a **zero vector 0** such that for every **u** in V, **u** + **0** = **u**.	Additive identity
5. For every **u** in V, there is a vector in V denoted by −**u** such that **u** + (−**u**) = **0**.	Additive inverse

Scalar Multiplication:

6. c**u** is in V.	Closure under scalar multiplication

Definition of a Vector Space (3 of 6)

Definition of a Vector Space

7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$	Distributive property
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$	Distributive property
9. $c(d\mathbf{u}) = (cd)\mathbf{u}$	Associative property
10. $1(\mathbf{u}) = \mathbf{u}$	Scalar identity

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Example 3 – The Vector Space of All 2 × 3 Matrices

Show that the set of all 2 × 3 matrices with the operations of matrix addition and scalar multiplication is a vector space.

Solution:

If A and B are 2 × 3 matrices and c is a scalar, then $A + B$ and cA are also 2 × 3 matrices. The set is, therefore, closed under matrix addition and scalar multiplication.

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Example 3 – Solution

Moreover, the other eight vector space axioms follow directly.

So, the set is a vector space. Vectors in this space have the form

$$\mathbf{a} = A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}.$$

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Example 5 – The Vector Space of continuous functions (calculus) (1 of 2)

Let $C(-\infty, \infty)$ be the set of all real-valued continuous functions defined on the entire real line. This set consists of all polynomial functions and all other continuous functions on the entire real line.

For example, *f*(*x*) = sin *x* and *g*(*x*) = e^x are members of this set.

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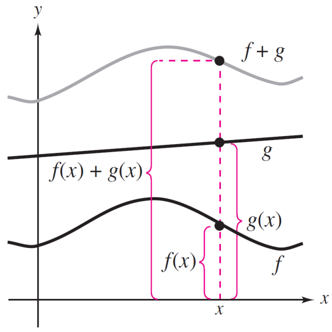
Example 5 – The Vector Space of continuous functions (calculus) (2 of 2)

Addition is defined by $(f + g)(x) = f(x) + g(x)$ as shown at the right.

Scalar multiplication is defined by $(cf)(x) = c[f(x)]$. Show that $C(-\infty, \infty)$ is a vector space.



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Example 5 – Solution (1 of 2)

To verify that the set $C(-\infty, \infty)$ is closed under addition and scalar multiplication, use a result from calculus—the sum of two continuous functions is continuous and the product of a scalar and a continuous function is continuous. To verify that the set $C(-\infty, \infty)$ has an additive identity, consider the function f_0 that has a value of zero for all x, that is,

$f_0(x) = 0$, x is any real number.

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Example 5 – Solution (2 of 2)

This function is continuous on the entire real line (its graph is simply the line $y = 0$), which means that it is in the set $C(-\infty, \infty)$. Moreover, if f is any other function that is continuous on the entire real line, then

$$(f + f_0)(x) = f(x) + f_0(x) = f(x) + 0 = f(x).$$

This shows that f_0 is the additive identity in $C(-\infty, \infty)$. The verification of the other vector space axioms is left to you.

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Definition of a Vector Space (4 of 6)

Summary of important Vector Spaces

R = set of all real numbers

R^2 = set of all ordered pairs

R^3 = set of all ordered triples

R^n = set of all n-tuples

$C(-\infty, \infty)$ = set of all continuous functions defined on the real number line

$C[a, b]$ = set of all continuous functions defined on a closed interval $[a, b]$, where $a \neq b$

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Definition of a Vector Space (5 of 6)

Summary of important Vector Spaces

P = set of all polynomials

P_n = set of all polynomials of degree $\leq n$
(together with the zero polynomial)

$M_{m,n}$ = set of all $m \times n$ matrices

$M_{n,n}$ = set of all $n \times n$ square matrices

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Definition of a Vector Space (6 of 6)

Theorem 4.4 properties of Scalar Multiplication

Let $\mathbf{v}$ be any element of a vector space V, and let c be any scalar. Then the properties below are true.

1. $0\mathbf{v} = \mathbf{0}$

2. $c\mathbf{0} = \mathbf{0}$

3. If $c\mathbf{v} = \mathbf{0}$, then $c = 0$ or $\mathbf{v} = \mathbf{0}$.

4. $(-1)\mathbf{v} = -\mathbf{v}$

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Sets That Are Not Vector Spaces

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Example 7 – The Set of Second-Degree Polynomials Is Not a Vector Space (1 of 1)

The set of all second-degree polynomials is not a vector space because it is not closed under addition. To see this, consider the second-degree polynomials $p(x) = x^2$ and $q(x) = 1 + x - x^2$ whose sum is the first-degree polynomial $p(x) + q(x) = 1 + x$.

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4 Vector Spaces

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4.3 Subspaces of Vector Spaces

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Objectives

- Determine whether a subset W of a vector space V is a subspace of V.
- Determine subspaces of R^n.

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Subspaces

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Subspaces (1 of 3)

Definition of a Subspace of a Vector Space

A nonempty subset W of a vector space V is a **subspace** of V when W is a vector space under the operations of addition and scalar multiplication defined in V.

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Example 1 – A Subspace of R^3

Show that the set $W = \{(x_1, 0, x_3)$: x_1 and x_3 are real numbers$\}$ is a subspace of R^3 with the standard operations.

Solution:

The set W is nonempty because it contains the zero vector (0, 0, 0).

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Example 1 – Solution (1 of 2)

Graphically, the set W can be interpreted as the xz-plane, as shown in Figure 4.7. The set W is closed under addition because the sum of any two vectors in the xz-plane must also lie in the xz-plane.

That is, if $(x_1, 0, x_3)$ and $(y_1, 0, y_3)$ are in W, then their sum $(x_1 + y_1, 0, x_3 + y_3)$ is also in W.

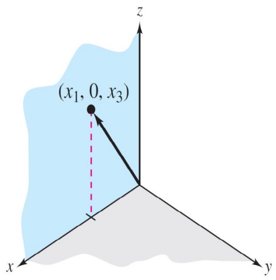

Figure 4.7

Example 1 – Solution (2 of 2)

Similarly, to see that W is closed under scalar multiplication, let $(x_1, 0, x_3)$ be in W and let c be a scalar.

Then $c(x_1, 0, x_3) = (cx_1, 0, cx_3)$ has zero as its second component and must be in W.

Subspaces (2 of 3)

Theorem 4.5 Test for a Subspace

If W is a nonempty subset of a vector space V, then W is a subspace of V if and only if the two closure conditions listed below hold.

1. If $\mathbf{u}$ and $\mathbf{v}$ are in W, then $\mathbf{u} + \mathbf{v}$ is in W.

2. If $\mathbf{u}$ is in W and c is any scalar, then $c\mathbf{u}$ is in W.

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Example 3 – The Set of Singular Matrices Is Not a Subspace of $M_{n,n}$

Let W be the set of singular matrices of order 2. Show that W is not a subspace of $M_{2,2}$ with the standard operations.

Solution:

By Test for a Subspace theorem, to show that a subset W is not a subspace, show that W is empty, W is not closed under addition, or W is not closed under scalar multiplication.

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Example 3 – Solution (1 of 2)

In this example, W is nonempty and closed under scalar multiplication, but it is not closed under addition. To see this, let A and B be

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Example 3 – Solution (2 of 2)

Then A and B are both singular (noninvertible), but their sum

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is nonsingular (invertible). So W is not closed under addition, and by Test for a Subspace theorem, W is not a subspace of $M_{2,2}$.

Subspaces (3 of 3)

Theorem 4.6 The Intersection of Two Subspaces Is a Subspace

If V and W are both subspaces of a vector space U, then the intersection of V and W (denoted by $V \cap W$) is also a subspace of U.

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Subspaces of R^n

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Subspaces of R^n (1 of 1)

R^n is a convenient source for examples of vector spaces, so the remainder of this section is devoted to looking at subspaces of R^n.

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Example 6 – Determining Subspaces of R^2

Determine whether each subset is a subspace of R^2.

a. The set of points on the line $x + 2y = 0$

b. The set of points on the line $x + 2y = 1$

Solution:

a. Solving for x, a point in R^2 is on the line $x + 2y = 0$ if and only if it has the form $(-2t, t)$, where t is any real number. (See Figure 4.10.)

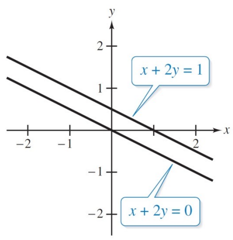

Figure 4.10

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Example 6 – Solution (1 of 2)

To show that this set is closed under addition, let $\mathbf{v}_1 = (-2t_1, t_1)$ and $\mathbf{v}_2 = (-2t_2, t_2)$ be any two points on the line. Then you have

$$\mathbf{v}_1 + \mathbf{v}_2 = (-2t_1, t_1) + (-2t_2, t_2)$$
$$= (-2(t_1 + t_2), t_1 + t_2) = (-2t_3, t_3)$$

where $t_3 = t_1 + t_2$. $\mathbf{v}_1 + \mathbf{v}_2$ lies on the line, and the set is closed under addition.

In a similar way, you can show that the set is closed under scalar multiplication. So, this set is a subspace of R^2.

Example 6 – Solution (2 of 2)

b. This subset of R^2 is *not* a subspace of R^2 because every subspace must contain the zero vector (0, 0), which is not on the line $x + 2y = 1$. (See Figure 4.10.)

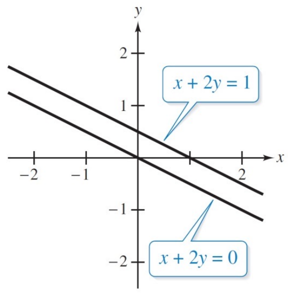

Figure 4.10

4 Vector Spaces

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4.4 Spanning Sets and Linear Independence

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Objectives

- Write a linear combination of a set of vectors in a vector space *V*.
- Determine whether a set *S* of vectors in a vector space *V* is a spanning set of *V*.
- Determine whether a set of vectors in a vector space *V* is linearly independent.

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Linear Combinations of Vectors in a Vector Space

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Linear Combinations of Vectors in a Vector Space (1 of 1)

Definition of a Linear Combination of Vectors

A vector $\mathbf{v}$ in a vector space V is a **linear combination** of the vectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k$ in V when $\mathbf{v}$ can be written in the form

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \ldots + c_k\mathbf{u}_k$$

where $c_1, c_2, \ldots, c_k$ are scalars.

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Example 2 – Finding a Linear Combination

Write the vector $\mathbf{w} = (1, 1, 1)$ as a linear combination of vectors in the set

$$S = \{\overset{\mathbf{v}_1}{(1, 2, 3)}, \overset{\mathbf{v}_2}{(0, 1, 2)}, \overset{\mathbf{v}_3}{(-1, 0, 1)}\}.$$

Solution:

Find scalars c_1, c_2, and c_3 such that

$$\begin{aligned}(1, 1, 1) &= c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-1, 0, 1)\\ &= (c_1, 2c_1, 3c_1) + (0, c_2, 2c_2) + (-c_3, 0, c_3)\\ &= (c_1 - c_3, 2c_1 + c_2, 3c_1 + 2c_2 + c_3).\end{aligned}$$

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Example 2 – Solution (1 of 2)

Equating corresponding components yields the system of linear equations below.

$$\begin{aligned} c_1 \qquad - c_3 &= 1 \\ 2c_1 + c_2 \qquad &= 1 \\ 3c_1 + 2c_2 + c_3 &= 1 \end{aligned}$$

Using Gauss-Jordan elimination, the augmented matrix of this system row reduces to

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

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Example 2 – Solution (2 of 2)

So, this system has infinitely many solutions, each of the form

$c_1 = 1 + t,\quad c_2 = -1 - 2t,\quad c_3 = t.$

To obtain one solution, you could let $t = 1$.

Then $c_3 = 1$, $c_2 = -3$, and $c_1 = 2$, and you have

$$\mathbf{w} = 2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3.$$

(Verify this.) Other choices for t would yield different ways to write **w** as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

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Spanning Sets

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Spanning Sets (1 of 3)

Definition of a Spanning Set of a Vector Space

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be a subset of a vector space V. The set S is a **spanning set** of V when *every* vector in V can be written as a linear combination of vectors in S. In such cases it is said that S **spans** V.

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Example 5 – A Spanning Set of R^3

Show that the set $S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$ spans R^3.

Solution:

Let $\mathbf{u} = (u_1, u_2, u_3)$ be *any* vector in R^3. Find scalars c_1, c_2, and c_3 such that

$$\begin{aligned}(u_1, u_2, u_3) &= c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-2, 0, 1)\\ &= (c_1 - 2c_3,\ 2c_1 + c_2,\ 3c_1 + 2c_2 + c_3).\end{aligned}$$

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Example 5 – Solution

This vector equation produces the system

$$\begin{aligned} c_1 \qquad - 2c_3 &= u_1\\ 2c_1 + c_2 \qquad\quad &= u_2\\ 3c_1 + 2c_2 + c_3 &= u_3.\end{aligned}$$

The coefficient matrix of this system has a nonzero. The system has a unique solution. So, any vector in R^3 can be written as a linear combination of the vectors in S, and you can conclude that the set S spans R^3.

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Spanning Set (2 of 3)

Definition of the Span of a Set

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a set of vectors in a vector space V, then the **span of S** is the set of all linear combinations of the vectors in S,

$\text{span}(S) = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_k\mathbf{v}_k : c_1, c_2, \ldots, c_k$ are real numbers$\}$.

The span of S is denoted by

$\text{span}(S)$ or $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$.

When $\text{span}(S) = V$, it is said that V is **spanned** by $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$, or that S **spans** V.

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Spanning Set (3 of 3)

Theorem 4.7 Span(S) is a Subspace of V

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a set of vectors in a vector space V, then $\text{span}(S)$ is a subspace of V.

Moreover, $\text{span}(S)$ is the smallest subspace of V that contains S, in the sense that every other subspace of V that contains S must contain $\text{span}(S)$.

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Linear Dependence and Linear Independence

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Linear Dependence and Linear Independence (1 of 5)

Definition of Linear Dependence and Linear Independence

A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ in a vector space V is **linearly independent** when the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_k\mathbf{v}_k = \mathbf{0}$$

has only the trivial solution

$$c_1 = 0, c_2 = 0, \ldots, c_k = 0.$$

If there are also nontrivial solutions, then S is **linearly dependent.**

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Example 7 – Examples of Linearly Dependent Sets

a. The set $S = \{(1, 2), (2, 4)\}$ in R^2 is linearly dependent because

$$-2(1, 2) + (2, 4) = (0, 0).$$

b. The set $S = \{(1, 0), (0, 1), (-2, 5)\}$ in R^2 is linearly dependent because

$$2(1, 0) - 5(0, 1) + (-2, 5) = (0, 0).$$

c. The set $S = \{(0, 0), (1, 2)\}$ in R^2 is linearly dependent because

$$1(0, 0) + 0(1, 2) = (0, 0).$$

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Linear Dependence and Linear Independence (2 of 5)

Testing for Linear Independence and Linear Dependence

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be a set of vectors in a vector space V. To determine whether S is linearly independent or linearly dependent, use the steps below.

1. From the vector equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_k\mathbf{v}_k = \mathbf{0}$, write a system of linear equations in the variables $c_1, c_2, \ldots,$ and c_k.

2. Determine whether the system has a unique solution.

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Linear Dependence and Linear Independence (3 of 5)

Testing for Linear Independence and Linear Dependence

3. If the system has only the trivial solution, $c_1 = 0$, $c_2 = 0$, . . . , $c_k = 0$, then the set S is linearly independent. If the system also has nontrivial solutions, then S is linearly dependent.

Example 9 – Testing for Linear Independence

Determine whether the set of vectors in P_2 is linearly independent or linearly dependent.

$$S = \{\underbrace{1 + x - 2x^2}_{\mathbf{v}_1}, \underbrace{2 + 5x - x^2}_{\mathbf{v}_2}, \underbrace{x + x^2}_{\mathbf{v}_3}\}$$

Solution:

Expanding the equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$ produces

$$c_1(1 + x - 2x^2) + c_2(2 + 5x - x^2) + c_3(x + x^2) = 0 + 0x + 0x^2$$
$$(c_1 + 2c_2) + (c_1 + 5c_2 + c_3)x + (-2c_1 - c_2 + c_3)x^2 = 0 + 0x + 0x^2.$$

Example 9 – Solution (1 of 2)

Equating corresponding coefficients of powers of x yields the homogeneous system of linear equations in c_1, c_2, and c_3 below.

$$\begin{aligned} c_1 + 2c_2 \quad\quad &= 0 \\ c_1 + 5c_2 + c_3 &= 0 \\ -2c_1 - c_2 + c_3 &= 0 \end{aligned}$$

The augmented matrix of this system reduces by Gaussian elimination as shown below.

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 5 & 1 & 0 \\ -2 & -1 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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Example 9 – Solution (2 of 2)

This implies that the system has infinitely many solutions. So, the system must have nontrivial solutions, and you can conclude that the set S is linearly dependent.

One nontrivial solution is

$c_1 = 2$, $c_2 = -1$, and $c_3 = 3$

which yields the nontrivial linear combination

$$(2)(1 + x - 2x^2) + (-1)(2 + 5x - x^2) + (3)(x + x^2) = 0.$$

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Linear Dependence and Linear Independence (4 of 5)

Theorem 4.8 A Property of Linearly Dependent Sets

A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$, $k \geq 2$, is linearly dependent if and only if at least one of the vectors $\mathbf{v}_i$ can be written as a linear combination of the other vectors in S.

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Example 12 – Writing a Vector as a Linear Combination of Other Vectors

In Example 9, you determined that the set

$$S = \{\overset{\mathbf{v}_1}{1 + x - 2x^2}, \overset{\mathbf{v}_2}{2 + 5x - x^2}, \overset{\mathbf{v}_3}{x + x^2}\}$$

is linearly dependent. Show that one of the vectors in this set can be written as a linear combination of the other two.

Solution:

In previous example, the equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$ produced the system

$$\begin{aligned} c_1 + 2c_2 &= 0 \\ c_1 + 5c_2 + c_3 &= 0 \\ -2c_1 - c_2 + c_3 &= 0. \end{aligned}$$

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Example 12 – Solution

This system has infinitely many solutions represented by $c_3 = 3t$, $c_2 = -t$, and $c_1 = 2t$. Letting $t = 1$ results in the equation $2\mathbf{v}_1 - \mathbf{v}_2 + 3\mathbf{v}_3 = \mathbf{0}$. So, $\mathbf{v}_2$ can be written as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_3$, as shown below.

$$\mathbf{v}_2 = 2\mathbf{v}_1 + 3\mathbf{v}_3$$

A check yields

$$2 + 5x - x^2 = 2(1 + x - 2x^2) + 3(x + x^2) = 2 + 5x - x^2.$$

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Linear Dependence and Linear Independence (5 of 5)

Theorem 4.8 Corollary

Two vectors **u** and **v** in a vector space *V* are linearly dependent if and only if one is a scalar multiple of the other.

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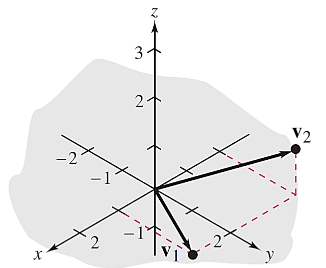
Example 13 – Testing for Linear Dependence of Two Vectors (1 of 2)

a. The set $S = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(1, 2, 0), (-2, 2, 1)\}$ is linearly independent because $\mathbf{v}_1$ and $\mathbf{v}_2$ are not scalar multiples of each other, as shown in Figure 4.15(a).



$S = \{(1, 2, 0), (-2, 2, 1)\}$
The set S is linearly independent.
Figure 4.15(a)

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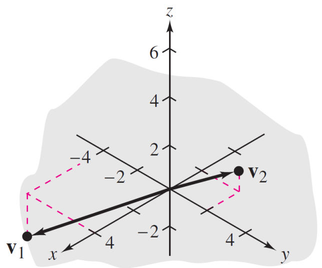
Example 13 – Testing for Linear Dependence of Two Vectors (2 of 2)

b. The set $S = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(4, -4, -2), (-2, 2, 1)\}$ is linearly dependent because $\mathbf{v}_1 = -2\mathbf{v}_2$, as shown in Figure 4.15(b).



$S = \{(4, -4, -2), (-2, 2, 1)\}$
The set S is linearly dependent.
Figure 4.15(b)

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4 Vector Spaces

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4.5 Basis and Dimension

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Objectives

- Recognize bases in the vector spaces R^n, P_n, and $M_{m,n}$.
- Find the dimension of a vector space.

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Basis for a Vector Space

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Basis for a Vector Space (1 of 4)

Definition of Basis

A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in a vector space V is a **basis** for V when the conditions below are true.

1. S spans V. **2.** S is linearly independent.

Moreover, if a vector space V has a basis with a finite number of vectors, then V is **finite dimensional.** Otherwise, V is **infinite dimensional.** The vector space $V = \{\mathbf{0}\}$, consisting of the zero vector alone, is finite dimensional.

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Example 1 – The Standard Basis for R^3

Show that the set below is a basis for R^3.

$S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

Solution:

In earlier example we showed that S spans R^3. Furthermore, S is linearly independent because the vector equation

$$c_1(1, 0, 0) + c_2(0, 1, 0) + c_3(0, 0, 1) = (0, 0, 0)$$

has only the trivial solution

$c_1 = c_2 = c_3 = 0$. (Verify this.)

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Example 1 – Solution

So, S is a basis for R^3. (See Figure 4.16.)

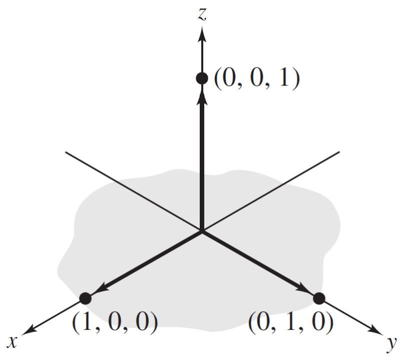

Figure 4.16

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Basis for a Vector Space (2 of 4)

Theorem 4.9 Uniqueness of Basis Representation

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for a vector space V, then every vector in V can be written in one and only one way as a linear combination of vectors in S.

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Example 6 – Uniqueness of Basis Representation

Let $\mathbf{u} = \{u_1, u_2, u_3\}$ be any vector in R^3.

Show that the equation $\mathbf{u} = c_1\,\mathbf{v}_1 + c_2\,\mathbf{v}_2 + c_3\,\mathbf{v}_3$ has a unique solution for the basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$.

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Example 6 – Solution (1 of 2)

From the equation

$$(u_1, u_2, u_3) = c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-2, 0, 1)$$
$$= (c_1 - 2c_3,\ 2c_1 + c_2,\ 3c_1 + 2c_2 + c_3)$$

you obtain the system of linear equations below.

$$\begin{aligned} c_1 \quad\quad\ \ - 2c_3 &= u_1 \\ 2c_1 + \ \ c_2 \quad\quad\ \ &= u_2 \\ 3c_1 + 2c_2 + \ \ c_3 &= u_3 \end{aligned}$$

$$\underbrace{\begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}}_{\mathbf{u}}$$

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Example 6 – Solution (2 of 2)

The matrix A is invertible, so you know this system has a unique solution, $\mathbf{c} = A^{-1}\mathbf{u}$.

Verify by finding A^{-1} that

$$c_1 = -u_1 + 4u_2 - 2u_3$$
$$c_2 = 2u_1 - 7u_2 + 4u_3$$
$$c_3 = -u_1 + 2u_2 - u_3.$$

For example, $\mathbf{u} = (1, 0, 0)$ can be represented uniquely as $-\mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3$.

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Basis for a Vector Space (3 of 4)

Theorem 4.10 Bases and Linear Dependence

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for a vector space V, then every set containing more than n vectors in V is linearly dependent.

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Example 7 – linearly Dependent sets in R^3 and P_3

a. R^3 has a basis consisting of three vectors, so the set

$$S = \{(1, 2, -1), (1, 1, 0), (2, 3, 0), (5, 9, -1)\}$$

must be linearly dependent.

b. P_3 has a basis consisting of four vectors, so the set

$$S = \{1, 1 + x, 1 - x, 1 + x + x^2, 1 - x + x^2\}$$

must be linearly dependent.

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Basis for a Vector Space (4 of 4)

Theorem 4.11 Number of Vectors in a Basis

If a vector space V has one basis with n vectors, then every basis for V has n vectors.

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Example 8 – Spanning Sets and Bases

Use Theorem 4.11 to explain why each statement is true.

a. The set $S_1 = \{(3, 2, 1), (7, -1, 4)\}$ is not a basis for R^3.

b. The set $S_2 = \{2 + x, x^2, -1 + x^3, 1 + 3x, 3 - 2x + x^2\}$ is not a basis for P_3.

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Example 8 – Solution

a. The standard basis for R^3, $S = \{(1, 0, 0), (0, 1, 0), (0,0,1)\}$, has three vectors, and S_1 has only two vectors. By Theorem 4.11, S_1 cannot be a basis for R^3.

b. The standard basis for P_3, $S = \{1, x, x^2, x^3\}$, has four vectors. By Theorem 4.11, the set S_2 has too many vectors to be a basis for P_3.

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The Dimension of a Vector Space

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The Dimension of a Vector Space (1 of 2)

Definition of the Dimension of a Vector Space

If a vector space V has a basis consisting of n vectors, then the number n is the **dimension** of V, denoted by $\dim(V) = n$. When V consists of the zero vector alone, the dimension of V is defined as zero.

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Example 9 – Finding Dimensions of Subspaces

Find the dimension of each subspace of R^3.

a. $W = \{(d, c - d, c)$: c and d are real numbers$\}$

b. $W = \{(2b, b, 0)$: b is a real number$\}$

Solution:

a. By writing the representative vector $(d, c - d, c)$ as

$(d, c - d, c) = (0, c, c) + (d, -d, 0) = c(0, 1, 1) + d(1, -1, 0)$

you can see that W is spanned by the set

$S = \{(0, 1, 1), (1, -1, 0)\}$.

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Example 9 – Solution

You can show that this set is linearly independent. So, S is a basis for W, and W is a two-dimensional subspace of R^3.

b. By writing the representative vector $(2b, b, 0)$ as $b(2, 1, 0)$, you can see that W is spanned by the set $S = \{(2, 1, 0)\}$. So, W is a one-dimensional subspace of R^3.

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The Dimension of a Vector Space (2 of 2)

Theorem 4.12 Basis Tests in an *n*-Dimensional Space

Let V be a vector space of dimension n.

1. If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a linearly independent set of vectors in V, then S is a basis for V.

2. If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ spans V, then S is a basis for V.

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Example 12 – Testing for a Basis in an *n*-Dimensional Space

Show that the set of vectors is a basis for $M_{5,1}$.

$$S = \left\{ \begin{array}{ccccc} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 & \mathbf{v}_5 \\ \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \\ 4 \end{bmatrix}, & \begin{bmatrix} 0 \\ 1 \\ 3 \\ -2 \\ 3 \end{bmatrix}, & \begin{bmatrix} 0 \\ 0 \\ 2 \\ -1 \\ 5 \end{bmatrix}, & \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2 \\ -3 \end{bmatrix}, & \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -2 \end{bmatrix} \end{array} \right\}$$

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Example 12 – Solution (1 of 2)

S has five vectors and the dimension of $M_{5,1}$ is 5, S is a basis by showing either that S is linearly independent or that S spans $M_{5,1}$.

To show that S is linearly independent, form the vector equation $c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 + c_5 v_5 = \mathbf{0}$, which yields the linear system below.

$$
\begin{aligned}
c_1 \qquad\qquad\qquad\qquad\qquad &= 0 \\
2c_1 + c_2 \qquad\qquad\qquad\qquad &= 0 \\
-c_1 + 3c_2 + 2c_3 \qquad\qquad\quad &= 0 \\
3c_1 - 2c_2 - c_3 + 2c_4 \qquad\; &= 0 \\
4c_1 + 3c_2 + 5c_3 - 3c_4 - 2c_5 &= 0
\end{aligned}
$$

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Example 12 – Solution (2 of 2)

This system has only the trivial solution, so S is linearly independent.

By Theorem 4.12, S is a basis for $M_{5,1}$.

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4 Vector Spaces

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4.6 Rank of a Matrix and Systems of Linear Equations

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Objectives

- Find a basis for the row space, a basis for the column space, and the rank of a matrix.
- Find the nullspace of a matrix.
- Find the solution of a consistent system $A\mathbf{x} = \mathbf{b}$ in the form $\mathbf{x}_p + \mathbf{x}_h$.

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Row Space, Column Space, and Rank of a Matrix

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Row Space, Column Space, and Rank of a Matrix (1 of 5)

In this section, you will investigate the vector space spanned by the row vectors (or column vectors) of a matrix.

For an $m \times n$ matrix A, we know that the n-tuples corresponding to the rows of A are the row vectors of A.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad \begin{array}{c} \text{Row Vectors of } A \\ (a_{11}, a_{12}, \ldots, a_{1n}) \\ (a_{21}, a_{22}, \ldots, a_{2n}) \\ \vdots \\ (a_{m1}, a_{m2}, \ldots, a_{mn}) \end{array}$$

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Row Space, Column Space, and Rank of a Matrix (2 of 5)

Similarly, the $m \times 1$ matrices corresponding to the columns of A are the column vectors of A.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad \begin{array}{c} \text{Column Vectors of } A \\ \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} \cdots \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} \end{array}$$

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Example 1 - Row Vectors and Column Vectors

For the matrix $A = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 3 & 4 \end{bmatrix}$, the row vectors are

(0, 1, −1) and (−2, 3, 4) and the column vectors are

$\begin{bmatrix} 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \text{and} \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$

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Row Space, Column Space, and Rank of a Matrix (3 of 5)

Definitions of Row Space and Column Space of a Matrix

Let *A* be an $m \times n$ matrix.

1. The **row space** of *A* is the subspace of R^n spanned by the row vectors of *A*.

2. The **column space** of *A* is the subspace of R^m spanned by the column vectors of *A*.

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Row Space, Column Space, and Rank of a Matrix (4 of 5)

Theorem 4.13 Row-Equivalent Matrices Have the Same Row Space

If an $m \times n$ matrix A is row-equivalent to an $m \times n$ matrix B, then the row space of A is equal to the row space of B.

Theorem 4.14 Basis for the Row Space of a Matrix

If a matrix A is row-equivalent to a matrix B in row-echelon form, then the nonzero row vectors of B form a basis for the row space of A.

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Example 2 - Finding a Basis for a Row Space

Find a basis for the row space of

$$A = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & -2 \end{bmatrix}.$$

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Example 2 - Solution

Using elementary row operations, rewrite *A* in row-echelon form as shown below.

$$B = \begin{array}{c} \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{l} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \mathbf{w}_3 \\ \\ \\ \end{array}$$

By Theorem 4.14, the nonzero row vectors of *B*, $\mathbf{w}_1 = (1, 3, 1, 3)$, $\mathbf{w}_2 = (0, 1, 1, 0)$, and $\mathbf{w}_3 = (0, 0, 0, 1)$, form a basis for the row space of *A*.

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Example 4 - Finding a Basis for the Column Space of a Matrix (Method 1)

Find a basis for the column space of matrix *A* from Example 2 by finding a basis for the row space of A^T.

Solution:

Write the transpose of *A* and use elementary row operations to write A^T in row-echelon form.

$$A^T = \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 3 & 1 & 0 & 4 & 0 \\ 1 & 1 & 6 & -2 & -4 \\ 3 & 0 & -1 & 1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -3 & 3 & 2 \\ 0 & 1 & 9 & -5 & -6 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \mathbf{w}_3 \\ \\ \end{array}$$

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Example 4 – Solution

So, $\mathbf{w}_1 = (1, 0, -3, 3, 2)$, $\mathbf{w}_2 = (0, 1, 9, -5, -6)$, and $\mathbf{w}_3 = (0, 0, 1, -1, -1)$ form a basis for the row space of A^T.

This is equivalent to saying that the column vectors $[1 \quad 0 \quad -3 \quad 3 \quad 2]^T$, $[0 \quad 1 \quad 9 \quad -5 \quad -6]^T$, and $[0 \quad 0 \quad 1 \quad -1 \quad -1]^T$ form a basis for the column space of A.

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Row Space, Column Space, and Rank of a Matrix (5 of 5)

Theorem 4.15 Row and Column Spaces Have Equal Dimensions

The row space and column space of an $m \times n$ matrix A have the same dimension.

Definition of the Rank of a Matrix

The dimension of the row (or column) space of a matrix A is the **rank** of A and is denoted by rank(A).

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Example 6 - Finding the Rank of a Matrix

To find the rank of the matrix *A* below, convert to a matrix *B* in row-echelon form as shown.

$$A = \begin{bmatrix} 1 & -2 & 0 & 1 \\ 2 & 1 & 5 & -3 \\ 0 & 1 & 3 & 5 \end{bmatrix} \rightarrow B = \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

The matrix *B* has three nonzero rows, so the rank of *A* is 3.

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The Nullspace of a Matrix

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The Nullspace of a Matrix (1 of 2)

THEOREM 4.16 Solutions of a Homogeneous System

If A is an $m \times n$ matrix, then the set of all solutions of the homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$ is a subspace of R^n called the **nullspace** of A and is denoted by $N(A)$. So,

$$N(A) = \{\mathbf{x} \in R^n: \ A\mathbf{x} = \mathbf{0}\}.$$

The dimension of the nullspace of A is the **nullity** of A.

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Example 7 - Finding the Nullspace of a Matrix

Find the nullspace of the matrix.

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 6 & -5 & 4 \\ 1 & 2 & 0 & 3 \end{bmatrix}$$

Solution:

The nullspace of A is the solution space of the homogeneous system

$$A\mathbf{x} = \mathbf{0}.$$

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Example 7 – Solution (1 of 3)

To solve this system, you could write the augmented matrix [A **0**] in reduced row-echelon form.

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 6 & -5 & 4 \\ 1 & 2 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Example 7 – Solution (2 of 3)

The system of equations corresponding to the reduced row-echelon form is

$$\begin{aligned} x_1 + 2x_2 \qquad + 3x_4 &= 0 \\ x_3 + \ \ x_4 &= 0 \end{aligned}$$

Choose x_2 and x_4 as free variables to represent the solutions in parametric form.

$$x_1 = -2s - 3t, \;\; x_2 = s, \;\; x_3 = -t, \;\; x_4 = t$$

Example 7 – Solution (3 of 3)

This means that the solution space of $A\mathbf{x} = \mathbf{0}$ consists of all solution vectors of the form

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s - 3t \\ s \\ -t \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

So, a basis for the nullspace of A consists of the vectors

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} -3 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

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The Nullspace of a Matrix (2 of 2)

Theorem 4.17 Dimension of the Solution Space

If A is an $m \times n$ matrix of rank r, then the dimension of the solution space of $A\mathbf{x} = \mathbf{0}$ is $n - r$. That is,
$n = \text{rank}(A) + \text{nullity}(A)$.

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Example 8 - Rank, nullity of a matrix, and Basis for the column Space

Let the column vectors of the matrix A be denoted by $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, $\mathbf{a}_4$, and $\mathbf{a}_5$. Find (a) the rank and nullity of A, and (b) a subset of the column vectors of A that forms a basis for the column space of A.

$$A = \begin{array}{c} \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix} \\ \begin{array}{ccccc} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 & \mathbf{a}_5 \end{array} \end{array}$$

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Example 8 – Solution (1 of 2)

Let B be the reduced row-echelon form of A.

$$A = \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix} \rightarrow B = \begin{bmatrix} 1 & 0 & -2 & 0 & 1 \\ 0 & 1 & 3 & 0 & -4 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

a. B has three nonzero rows, so the rank of A is 3. Also, the number of columns of A is $n = 5$, which implies that the nullity of A is $n -$ rank $= 5 - 3 = 2$.

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Example 8 – Solution (2 of 2)

b. The first, second, and fourth column vectors of *B* are linearly independent, so the corresponding column vectors of *A*,

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 3 \end{bmatrix}, \quad \text{and} \quad \mathbf{a}_4 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

form a basis for the column space of *A*.

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Solutions of Systems of Linear Equations

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Solutions of Systems of Linear Equations (1 of 4)

Theorem 4.18 Solutions of a Nonhomogeneous Linear System

If $\mathbf{x}_p$ is a particular solution of the nonhomogeneous system $A\mathbf{x} = \mathbf{b}$, then every solution of this system can be written in the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$, where $\mathbf{x}_h$ is a solution of the corresponding homogeneous system $A\mathbf{x} = \mathbf{0}$.

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Example 9 - Finding the Solution Set of a Nonhomogeneous System

Find the set of all solution vectors of the system of linear equations.

$$\begin{aligned} x_1 \qquad\quad - 2x_3 + x_4 &= 5 \\ 3x_1 + x_2 - 5x_3 \qquad &= 8 \\ x_1 + 2x_2 \qquad - 5x_4 &= -9 \end{aligned}$$

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Example 9 – Solution (1 of 2)

The augmented matrix for the system $A\mathbf{x} = \mathbf{b}$ reduces as shown below.

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 5 \\ 3 & 1 & -5 & 0 & 8 \\ 1 & 2 & 0 & -5 & -9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 & 1 & 5 \\ 0 & 1 & 1 & -3 & -7 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The system of linear equations corresponding to the reduced row-echelon matrix is

$$\begin{aligned} x_1 \quad\quad - 2x_3 + \quad x_4 &= 5 \\ x_2 + \quad x_3 - 3x_4 &= -7 \end{aligned}$$

Example 9 – Solution (2 of 2)

Letting $x_3 = s$ and $x_4 = t$, write a representative solution vector of $A\mathbf{x} = \mathbf{b}$ as shown below.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 + 2s - t \\ -7 - s + 3t \\ 0 + s + 0t \\ 0 + 0s + t \end{bmatrix} = \begin{bmatrix} 5 \\ -7 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 3 \\ 0 \\ 1 \end{bmatrix} = \mathbf{x}_p + s\mathbf{u}_1 + t\mathbf{u}_2$$

$\mathbf{x}_p$ is a *particular* solution vector of $A\mathbf{x} = \mathbf{b}$, and $\mathbf{x}_h = s\mathbf{u}_1 + t\mathbf{u}_2$ represents an arbitrary vector in the solution space of $A\mathbf{x} = \mathbf{0}$.

Solutions of Systems of Linear Equations (2 of 4)

Theorem 4.19 Solutions of a System of Linear Equations

The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A.

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Example 10 - Consistency of a System of Linear Equations (1 of 2)

Consider the system of linear equations

$$\begin{aligned} x_1 + x_2 - x_3 &= -1 \\ x_1 \qquad + x_3 &= 3 \\ 3x_1 + 2x_2 - x_3 &= -9 \end{aligned}$$

The augmented matrix for the system is

$$[A \quad \mathbf{b}] = \begin{array}{c} \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 0 & 1 & 3 \\ 3 & 2 & -1 & 1 \end{bmatrix}. \\ \begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{b} \end{array} \end{array}$$

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Example 10 - Consistency of a System of Linear Equations (2 of 2)

Notice that $\mathbf{b} = 2\mathbf{a}_1 - 2\mathbf{a}_2 + \mathbf{a}_3$.

So, **b** is in the column space of *A*, and the system of linear equations is consistent.

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Solutions of Systems of Linear Equations (3 of 4)

Summary of Equivalent Conditions for Square Matrices

If A is an $n \times n$ matrix, then the conditions below are equivalent.

1. A is invertible.

2. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $n \times 1$ matrix $\mathbf{b}$.

3. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

4. A is row-equivalent to I_n.

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Solutions of Systems of Linear Equations (4 of 4)

Summary of Equivalent Conditions for Square Matrices

5. $A \neq 0$

6. Rank(A) = n

7. The n row vectors of A are linearly independent.

8. The n column vectors of A are linearly independent.

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4 Vector Spaces

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4.7 Coordinates and Change of Basis

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Objectives

- Find a coordinate matrix relative to a basis in R^n.
- Find the transition matrix from the basis B to the basis B in R^n.
- Represent coordinates in general n-dimensional spaces.

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Coordinate Representation in R^n

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Coordinate Representation in R^n of 3)

If B is a basis for a vector space V, then every vector **x** in V can be expressed in one and only one way as a linear combination of vectors in B. The coefficients in the linear combination are the **coordinates of x relative to *B*.**

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Coordinate Representation in R^n (2 of 3)

Coordinate Representation Relative to a Basis

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be an ordered basis for a vector space V and let $\mathbf{x}$ be a vector in V such that

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n.$$

The scalars $c_1, c_2, \ldots, c_n$ are the **coordinates of x relative to the basis *B*.**

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Coordinate Representation in R^n (3 of 3)

Coordinate Representation Relative to a Basis

The **coordinate matrix** (or **coordinate vector**) **of x relative to *B*** is the column matrix in R^n whose components are the coordinates of **x**.

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

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Example 1 – Coordinates and Components in R^n

Find the coordinate matrix of **x** = (−2, 1, 3) in R^3 relative to the standard basis

S = {(1, 0, 0), (0, 1, 0), (0, 0, 1)}.

Solution:

The vector **x** can be written as

x = (−2, 1, 3) = −2(1, 0, 0) + 1(0, 1, 0) + 3(0, 0, 1),

so the coordinate matrix of **x** relative to the standard basis is simply

$$[\mathbf{x}]_S = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}.$$

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Example 1 – Solution

The components of **x** are the same as its coordinates relative to the standard basis.

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Example 2 – Finding a Coordinate Matrix Relative to a Standard Basis

The coordinate matrix of **x** in R^2 relative to the (nonstandard) ordered basis $B = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(1, 0), (1, 2)\}$ is

$$[\mathbf{x}]_B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Find the coordinate matrix of **x** relative to the standard basis $B' = \{\mathbf{u}_1, \mathbf{u}_2\} = \{(1, 0), (0, 1)\}$.

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Example 2 – Solution (1 of 2)

The coordinate matrix of **x** relative to B is $[\mathbf{x}]_B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, so

$\mathbf{x} = 3\mathbf{v}_1 + 2\mathbf{v}_2 = 3(1, 0) + 2(1, 2) = (5, 4) = 5(1, 0) + 4(0, 1).$

It follows that the coordinate matrix of **x** relative to B' is

$$[\mathbf{x}]_{B'} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}.$$

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Example 2 – Solution (2 of 2)

Figure 4.17 compares these two coordinate representations.

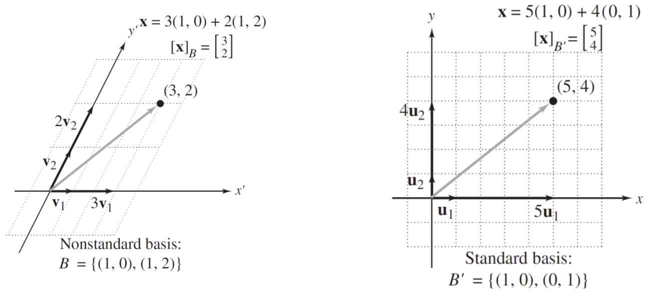

Figure 4.17

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Change of Basis in R^n

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Change of Basis in R^n (1 of 7)

The procedure demonstrated in Examples 2 and 3 is called a **change of basis.** That is, you were given the coordinates of a vector relative to a basis *B* and were asked to find the coordinates relative to another basis *B′*.

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Change of Basis in R^n (2 of 7)

Assume that

$$B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} \text{ and } B' = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$$

are two ordered bases for R^n. If **x** is a vector in R^n and

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \quad \text{and} \quad [\mathbf{x}]_{B'} = \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{bmatrix}$$

are the coordinate matrices of **x** relative to *B* and *B′*, then the **transition matrix *P* from *B′* to *B*** is the matrix *P* such that

$$[\mathbf{x}]_B = P[\mathbf{x}]_{B'}.$$

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Change of Basis in R^n (3 of 7)

The next theorem tells you that the transition matrix P is invertible and its inverse is the **transition matrix from *B* to *B*′**. That is,

$$[\mathbf{x}]_{B'} = P^{-1}[\mathbf{x}]_B.$$

Coordinate matrix of $\mathbf{x}$ relative to B'

Transition matrix from B to B'

Coordinate matrix of $\mathbf{x}$ relative to B

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Change of Basis in R^n (4 of 7)

Theorem 4.20 The Inverse of a Transition Matrix

If P is the transition matrix from a basis B' to a basis B in R^n, then P is invertible and the transition matrix from B to B' is P^{-1}.

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Change of Basis in R^n (5 of 7)

LEMMA

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $B' = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be two bases for a vector space V. If

$$\begin{aligned}
\mathbf{v}_1 &= c_{11}\mathbf{u}_1 + c_{21}\mathbf{u}_2 + \cdots + c_{n1}\mathbf{u}_n \\
\mathbf{v}_2 &= c_{12}\mathbf{u}_1 + c_{22}\mathbf{u}_2 + \cdots + c_{n2}\mathbf{u}_n \\
&\vdots \\
\mathbf{v}_n &= c_{1n}\mathbf{u}_1 + c_{2n}\mathbf{u}_2 + \cdots + c_{nn}\mathbf{u}_n
\end{aligned}$$

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Change of Basis in R^n (6 of 7)

LEMMA

Then the transition matrix from B to B' is

$$Q = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix}.$$

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Change of Basis in R^n (7 of 7)

Theorem 4.21 Transition Matrix from B to B'

Let

$B = \{\mathbf{v}_1, \mathbf{v}_2, . . . , \mathbf{v}_n\}$ and $B' = \{\mathbf{u}_1, \mathbf{u}_2, . . . , \mathbf{u}_n\}$

be two bases for R^n. Then the transition matrix P^{-1} from B to B' can be found by using Gauss-Jordan elimination on the $n \times 2n$ matrix $[B' \;\; B]$, as shown below.

$$[B' \quad B] \rightarrow [I_n \quad P^{-1}]$$

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Example 4 – Finding a transition Matrix

Find the transition matrix from B to B' for the bases for R^3 below.

$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and

$B' = \{(1, 0, 1), (0, -1, 2), (2, 3, -5)\}$

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Example 4 – Solution (1 of 2)

First use the vectors in the two bases to form the matrices B and B'.

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B' = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 3 \\ 1 & 2 & -5 \end{bmatrix}$$

Then form the matrix $[B' \;\; B]$ and use Gauss-Jordan elimination to rewrite $[B' \;\; B]$ as $[I_3 \quad P^{-1}]$.

$$\left[\begin{array}{rrrrrr} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 1 & 2 & -5 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{rrrrrr} 1 & 0 & 0 & -1 & 4 & 2 \\ 0 & 1 & 0 & 3 & -7 & -3 \\ 0 & 0 & 1 & 1 & -2 & -1 \end{array}\right]$$

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Example 4 – Solution (2 of 2)

From this, you can conclude that the transition matrix from B to B'

$$P^{-1} = \left[\begin{array}{rrr} -1 & 4 & 2 \\ 3 & -7 & -3 \\ 1 & -2 & -1 \end{array}\right].$$

Multiply P^{-1} by the coordinate matrix of $\mathbf{x} = [1 \quad 2 \quad -1]^T$.

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Coordinate Representation in General n-Dimensional Spaces

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Example 6 – Coordinate Representation in P_3

Find the coordinate matrix of

$$p = 4 - 2x^2 + 3x^3$$

relative to the standard basis for P_3,

$$S = \{1, x, x^2, x^3\}.$$

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Example 6 – Solution

Write p as a linear combination of the basis vectors (in the given order).

$$p = 4(1) + 0(x) + (-2)(x^2) + 3(x^3)$$

So, the coordinate matrix of p relative to S is

$$[p]_S = \begin{bmatrix} 4 \\ 0 \\ -2 \\ 3 \end{bmatrix}.$$

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4 Vector Spaces

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4.8 Applications of Vector Spaces

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Objectives

- Use the Wronskian to test a set of solutions of a linear homogeneous differential equation for linear independence.
- Identify and sketch the graph of a conic section and perform a rotation of axes.

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Linear Differential Equations (Calculus)

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Linear Differential Equations (Calculus) (1 of 4)

A **linear differential equation of order *n*** is of the form

$$y^{(n)} + g_{n-1}(x)y^{(n-1)} + \cdot \cdot \cdot + g_1(x)y' + g_0(x)y = f(x)$$

where g_0, g_1, . . . , g_{n-1} and f are functions of x with a common domain. If $f(x) = 0$, then the equation is **homogeneous.**

Otherwise it is **nonhomogeneous.** A function y is a **solution** of the linear differential equation if the equation is satisfied when y and its first n derivatives are substituted into the equation.

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Example 1 – A Second-Order Linear Differential Equation

Show that both $y = e^x$ and $y_2 = e^{-x}$ are solutions of the second-order linear differential equation $y'' - y = 0$.

Solution:

For the function $y_1 = e^x$, you have $y_1' = e^x$ and $y_1'' = e^x$.

So,

$$y_1'' - y_1 = e^x - e^x = 0$$

which means that $y_1 = e^x$ is a solution of the differential equation. Similarly, for $y_2 = e^{-x}$, you have

$$y_2' = -e^{-x} \quad \text{and} \quad y_2'' = e^{-x}.$$

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Example 1 – Solution

This implies that

$$y_2'' - y_2 = e^{-x} - e^{-x} = 0.$$

So, $y_2 = e^{-x}$ is also a solution of the linear differential equation.

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Linear Differential Equations (Calculus) (2 of 4)

Solutions of a Linear Homogeneous Differential Equation

Every nth-order linear homogeneous differential equation

$$y^{(n)} + g_{n-1}(x)y^{(n-1)} + \cdot \cdot \cdot + g_1(x)y' + g_0(x)y = 0$$

has n linearly independent solutions. Moreover, if $\{y_1, y_2, . . . , y_n\}$ is a set of linearly independent solutions, then every solution is of the form

$y = C_1y_1 + C_2y_2 + . . . + C_ny_n$

where $C_1 , C_2 , . . . , C_n$ are real numbers.

Linear Differential Equations (Calculus) (3 of 4)

Definition of the Wronskian of a Set of Functions

Let $\{y_1, y_2, . . . , y_n\}$ be a set of functions, each of which has $n - 1$ derivatives on an interval I. The determinant

$$W(y_1, y_2, . . . , y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

is the **Wronskian** of the set of functions.

Example 2 – Finding the Wronskian of a Set of Functions

a. The Wronskian of the set $\{1 - x, 1 + x, 2 - x\}$ is

$$W = \begin{vmatrix} 1 - x & 1 + x & 2 - x \\ -1 & 1 & -1 \\ 0 & 0 & 0 \end{vmatrix} = 0.$$

b. The Wronskian of the set $\{x, x^2, x^3\}$ is

$$W = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} = 2x^3.$$

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Linear Differential Equations (Calculus) (4 of 4)

Wronskian Test for Linear Independence

Let $\{y_1, y_2, \ldots, y_n\}$ be a set of n solutions of an nth-order linear homogeneous differential equation. This set is linearly independent if and only if the Wronskian is not identically equal to zero.

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Example 3 – Testing a Set of Solutions for Linear Independence

Determine whether {1, cos *x*, sin *x*} is a set of linearly independent solutions of the linear homogeneous differential equation

$$y''' + y' = 0.$$

Solution:

Begin by observing that each of the functions is a solution of $y''' + y' = 0.$

Example 3 – Solution (1 of 2)

Next, testing for linear independence produces the Wronskian of the three functions, as shown below.

$$W = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix}$$
$$= \sin^2 x + \cos^2 x$$
$$= 1$$

The Wronskian *W* is not identically equal to zero, so the set {1, cos *x*, sin *x*} is linearly independent.

Example 3 – Solution (2 of 2)

Moreover, this set consists of three linearly independent solutions of a third-order linear homogeneous differential equation, so the general solution is

$y = C_1 + C_2 \cos x + C_3 \sin x$

where C_1 , C_2, and C_3 are real numbers.

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Conic Sections and Rotation

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Conic Sections and Rotation (1 of 5)

Every conic section in the *xy*-plane has an equation that can be written in the form

$$ax^2 + bxy + cy^2 + dx + ey + f = 0.$$

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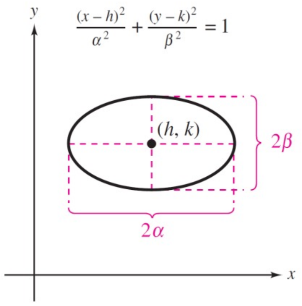
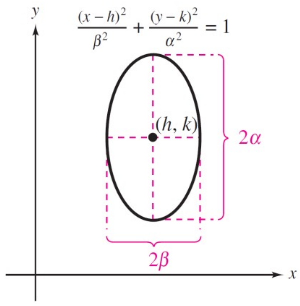
Conic Sections and Rotation (2 of 5)

Standard Forms of Equations of Conics

Circle (r = radius): $(x - h)^2 + (y - k)^2 = r^2$

Ellipse (2α = major axis length, 2β = minor axis length):





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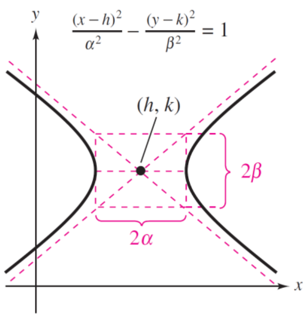
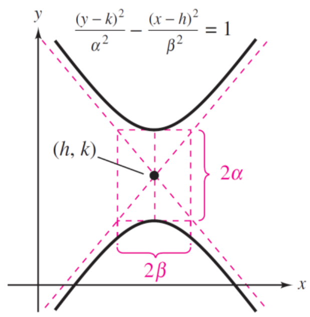
Conic Sections and Rotation (3 of 5)

Standard Forms of Equations of Conics

Hyperbola (2α = transverse axis length, 2β = conjugate axis length):





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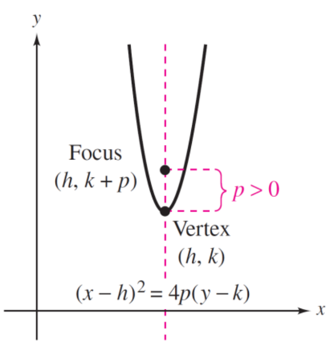
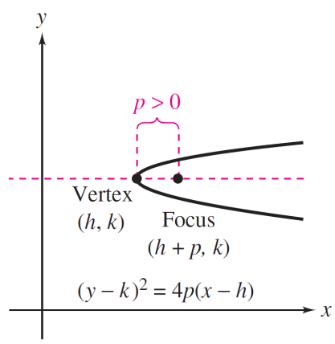
Conic Sections and Rotation (4 of 5)

Standard Forms of Equations of Conics

Parabola (p = directed distance from vertex to focus):





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Example 5 – Identifying Conic Sections (1 of 4)

a. The standard form of $x^2 - 2x + 4y - 3 = 0$ is

$$(x - 1)^2 = 4(-1)(y - 1).$$

The graph of this equation is a parabola with the vertex at $(h, k) = (1, 1)$. The axis of the parabola is vertical. The directed distance p from the vertex to the focus is $p = -1$, so the focus is the point $(1, 0)$.

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Example 5 – Identifying Conic Sections (2 of 4)

Finally, the focus lies below the vertex, so the parabola opens downward, as shown in Figure 4.18(a).

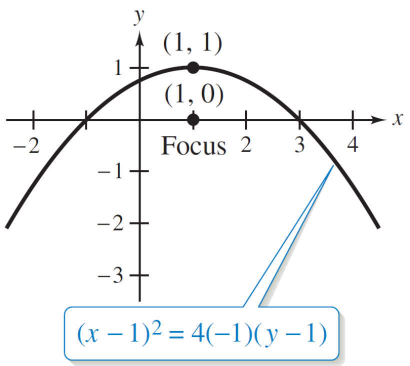

Figure 4.18(a)

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Example 5 – Identifying Conic Sections (3 of 4)

b. The standard form of $x^2 + 4y^2 + 6x - 8y + 9 = 0$ is

$$\frac{(x+3)^2}{4} + \frac{(y-1)^2}{1} = 1.$$

The graph of this equation is an ellipse with its centre at $(h, k) = (-3, 1)$.

The major axis is horizontal, and its length is $2\alpha = 4$. The length of the minor axis is $2\beta = 2$.

Example 5 – Identifying Conic Sections (4 of 4)

The vertices of this ellipse occur at $(-5, 1)$ and $(-1, 1)$, and the endpoints of the minor axis occur at $(-3, 2)$ and $(-3, 0)$, as in Figure 4.18(b).

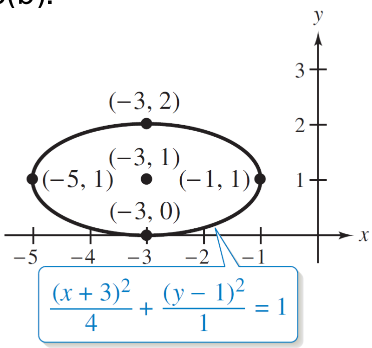

Figure 4.18(b)

Conic Sections and Rotation (5 of 5)

Rotation of axes

The general second-degree equation

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

can be written in the form

$$a'(x')^2 + c'(y')^2 + d'x' + e'y' + f' = 0$$

by rotating the coordinate axes counterclockwise through the angle θ, where θ is found using the equation $\cot 2\theta = (a - c)/b$. The coefficients of the new equation are obtained from the substitutions

$$x = x' \cos \theta - y' \sin \theta \text{ and } y = x' \sin \theta + y' \cos \theta.$$

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Example 7 – Rotation of a Conic Section

Perform a rotation of axes to eliminate the *xy*-term in

$$5x^2 - 6xy + 5y^2 + 14\sqrt{2}x - 2\sqrt{2}y + 18 = 0$$

and sketch the graph of the resulting equation in the $x'y'$-plane.

Solution:

Find the angle of rotation θ using

$$\cot 2\theta = \frac{a - c}{b} = \frac{5 - 5}{-6} = 0.$$

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Example 7 – Solution (1 of 3)

This implies that $\theta = \pi/4$. So,

$$\sin \theta = \frac{1}{\sqrt{2}} \quad \text{and} \quad \cos \theta = \frac{1}{\sqrt{2}}.$$

By substituting

$$x = x' \cos \theta - y' \sin \theta = \frac{1}{\sqrt{2}}(x' - y')$$

and

$$y = x' \sin \theta + y' \cos \theta = \frac{1}{\sqrt{2}}(x' + y')$$

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Example 7 – Solution (2 of 3)

Into the original equation and simplifying, verify that you obtain

$$(x')^2 + 4(y')^2 + 6x' - 8y' + 9 = 0.$$

Finally, by completing the square, the standard form of this equation is

$$\frac{(x' + 3)^2}{2^2} + \frac{(y' - 1)^2}{1^2} = \frac{(x' + 3)^2}{4} + \frac{(y' - 1)^2}{1} = 1$$

which is the equation of an ellipse.

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Example 7 – Solution (3 of 3)

As shown in Figure 4.19.

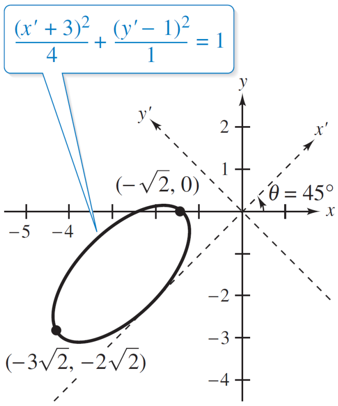

Figure 4.19

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